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CS441 Finalterm mcqs

Which of the following is not considered a core characteristic of Big Data?

A) Volume

B) Variety

C) Velocity

D) Value

The term "Big Data" gained popularity in 2001 with the introduction of the 3Vs model by:

A) Doug Laney

B) Tim Berners-Lee

C) Vint Cerf

D) Jeff Bezos

Which component of Hadoop is responsible for storing metadata about files and directories, but not the actual data?

A) DataNode

B) NameNode

C) ResourceManager

D) Secondary NameNode

In Hadoop Distributed File System (HDFS), a large file is split into fixed-size chunks called:

A) Segments

B) Packets

C) Records

D) Blocks

What is the default block size for a file in modern versions of HDFS?

A) 64 MB

B) 32 MB

C) 128 MB

D) 256 MB

Which of the following is a primary reason for data replication in HDFS?

A) To speed up data processing

B) To reduce storage space

C) To enable real-time updates

D) To ensure fault tolerance

The processing layer of Hadoop that follows a distributed programming model is known as:

A) Hive

B) MapReduce

C) HDFS

D) Spark

In the MapReduce workflow, which phase is responsible for grouping intermediate key-value pairs?

A) Mapping

B) Reducing

C) Input Splitting

D) Shuffling & Sorting

Which component of the Hadoop ecosystem acts as a resource manager and schedules jobs across the cluster?

A) HDFS

B) MapReduce

C) YARN

D) Hive

What is the primary purpose of a Combiner in the MapReduce process?

A) To write final output to HDFS

B) To execute the Reduce function

C) To split input data into chunks

D) To perform local aggregation to reduce network traffic

Hadoop is primarily designed for which of the following workloads?

A) Real-time transaction processing

B) Online analytical processing (OLAP)

C) Batch processing and large-scale analytics

D) Random read/write operations

Which of the following describes the "Schema-on-Read" approach used by Hive?

A) Data must conform to a fixed schema before being written.

B) The schema is applied after the data is processed.

C) The schema is not required at all.

D) The schema is applied when the data is read (queried).

A Hive table whose data is stored in a user-specified HDFS location and is not deleted when the table is dropped is called a(n):

A) Managed table

B) Partitioned table

C) External table

D) Bucketed table

What is the core difference between the WHERE and HAVING clauses in HiveQL?

A) HAVING is used for joining tables, while WHERE is for filtering.

B) WHERE filters data before MapReduce, while HAVING filters after.

C) WHERE filters rows before grouping, while HAVING filters after aggregation.

D) They are interchangeable and serve the same purpose.

Which of the following is not a common aggregate function used in HiveQL?

A) SUM()

B) AVG()

C) COUNT()

D) LEAD()

The HiveQL query SELECT department, SUM(salary) FROM employees GROUP BY department HAVING SUM(salary) > 100000; would be translated by Hive into:

A) A single SQL query

B) One or more MapReduce jobs

C) A real-time database transaction

D) A Python script

Which of the following operations is Hive generally not suited for?

A) Data summarization

B) ETL (Extract, Transform, Load) operations

C) Real-time transactional updates

D) Data warehousing

In the context of the Hadoop ecosystem, what is the role of a DataNode?

A) To store metadata about files

B) To store the actual data blocks

C) To manage user permissions

D) To handle job scheduling

The ROW_NUMBER() window function in HiveQL is used to:

A) Calculate a cumulative sum

B) Filter rows based on a condition

C) Assign a unique, sequential number to rows within a partition

D) Get the value from a preceding row

Which of the following is a primary limitation of using HDFS for data storage?

A) It is not fault-tolerant.

B) It requires expensive, specialized hardware.

C) It is not scalable.

D) It is inefficient for storing a large number of small files.

What is the purpose of Apache Flume in the Hadoop ecosystem?

A) To schedule jobs and workflows

B) To provide a NoSQL database

C) To transfer data from RDBMS to HDFS

D) To ingest streaming data like logs and events

Which component of Hadoop is primarily responsible for fault tolerance?

A) The client

B) The application layer

C) The NameNode and DataNodes working together

D) The MapReduce job scheduler

The primary use case for Apache Sqoop is:

A) Querying data with HiveQL

B) Running machine learning models

C) Transferring data between HDFS and relational databases

D) Coordinating distributed services

Which of the following is a characteristic of managed tables in Hive?

A) The data files remain after the table is dropped.

B) The data is not stored in HDFS.

C) Both the data and metadata are managed by Hive.

D) They are primarily used for real-time transactions.

What is the key difference in latency between Hive and a traditional RDBMS?

A) RDBMS has high latency for large datasets; Hive has low latency.

B) Both have similar, low latency.

C) Both have similar, high latency.

D) RDBMS has low latency; Hive has high latency due to batch processing.

The LOCATION clause in a Hive table definition is specifically used for:

A) Managed tables

B) External tables

C) Temporary tables

D) Partitioned tables

In the Hadoop architecture, what layer sits between the processing layer and HDFS?

A) Hive

B) MapReduce

C) YARN

D) HBase

Which of the following is a key advantage of using Hive over writing raw MapReduce code in Java?

A) It has better performance for small datasets.

B) It allows users with SQL knowledge to query data easily.

C) It supports real-time data updates.

D) It provides a graphical user interface for data visualization.

What is the name of the central service in Hadoop that is used for distributed application coordination?

A) YARN

B) Oozie

C) Spark

D) ZooKeeper

The WHERE clause in HiveQL is primarily executed during which phase of the query?

A) The Reducer phase

B) The final output phase

C) The Mapper phase (before aggregation)

D) The Join phase

The LIKE operator in HiveQL is used for:

A) Exact string matching

B) Pattern matching in strings

C) Comparing two different data types

D) Performing a join operation

A language is called Turing-decidable if:

A) A Turing Machine can accept all strings in the language.

B) A Turing Machine can halt on every input, accepting or rejecting.

C) A Turing Machine can accept all strings but may not halt on others.

D) It is an undecidable language.

Which of the following is true about Turing-recognizable languages?

A) They are a subset of Turing-decidable languages.

B) They are also known as recursive languages.

C) A Turing Machine will halt and accept all strings in the language.

D) A Turing Machine is guaranteed to halt on all inputs.

The set of Context-Free Languages is closed under which of the following operations?

A) Intersection

B) Complement

C) Set Difference

D) Union

Which class of languages is recognized by a Finite Automaton (FA)?

A) Context-Free Languages

B) Context-Sensitive Languages

C) Regular Languages

D) Recursively Enumerable Languages

A language that is Turing-recognizable but not Turing-decidable is:

A) A regular language

B) A context-free language

C) A context-sensitive language

D) A non-recursive language

The closure property of Recursive Languages is:

A) Closed under Union and Concatenation, but not Complement.

B) Closed under all operations except Intersection.

C) Closed under all common operations (Union, Intersection, Complement, etc.).

D) Not closed under any operation.

The language $L=\{a^n b^n c^n \mid n \geq 0\}$ is an example of a language that is:

A) Regular

B) Context-Free

C) Not Context-Free

D) Turing-decidable

If a language is decidable, then its complement is also:

A) Recognizable

B) Regular

C) Decidable

D) Undecidable

The class of languages that is not closed under the complement operation is:

A) Regular Languages

B) Context-Free Languages

C) Recursive Languages

D) Recursively Enumerable Languages

Which of the following is a key difference between a Pushdown Automaton (PDA) and a Finite Automaton (FA)?

A) A PDA has a finite number of states.

B) A PDA uses a stack for memory.

C) A PDA can accept more powerful languages.

D) A PDA is less powerful than an FA.

The Halting Problem for Turing Machines is a classic example of a problem that is:

A) Decidable

B) Undecidable

C) Recognizable

D) Regular